Supplementary Material for HKVM-RAG: Key-Value-Separated Hypergraph Evidence Organization for Multi-Hop RAG

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I. Evidence Map and Claim Boundaries

This supplementary material is organized as an audit trail for the main paper's claims. It follows the paper's bounded framing: HKVM-RAG is evaluated as a fixed-substrate evidence-indexing and dense-aware evidence-control method, not as a hidden-test benchmark, a standalone dense-retrieval replacement, or a full reproduction of every related RAG system. All tables below are derived from frozen prediction files and claim-specific result files. No supplementary analysis introduces new LLM extraction calls, new dense-retrieval runs, or new answer-generation calls.

II. Fixed-Substrate Protocol and Artifact Manifests

The fixed substrate is both a scientific control and an evaluation boundary. Structured methods share the same candidate-passage pools, deterministic DeepSeek evidence-tuple extraction schema, cached passage-level tuple files, top-10 retrieval budget, top-5 answer-context budget, train/dev splits, and seeds. The reported passage-unit counts are cached extractor-input counts, not retrieval budgets: 2WikiMultiHopQA, MuSiQue, and HotpotQA expose different numbers of candidate passages before the shared retrieval and answer-context budgets are applied. Changing the extractor endpoint, prompt, schema, tuple budget, candidate-passage source, or answer reader creates a new substrate and requires a new manifest rather than a direct extension of the frozen tables.

The paper-facing dev structured matrix uses 202,176 cached dev passage-level extractor inputs: 125,760 for 2WikiMultiHopQA, 48,325 for MuSiQue, and 28,091 for HotpotQA. Train-side controller inputs use 399,491 additional cached passage-level extractor inputs. These train-side caches are used only for the train-derived support scorer, learned HKVM calibration, dense-aware controller, and transfer/calibration audits. Gold support is not sent to the extractor; support labels are used only after cache construction for supervised ranker/controller training.

The released artifact records benchmark data, extraction caches, frozen prediction outputs, compact result summaries, SHA-256 file manifests, reproduction scripts, run summaries, and per-family provenance records. Dense model weights and API credentials are not redistributed. Rerunning dense retrieval requires downloading the upstream model weights, and regenerating LLM evidence tuples requires a new extraction manifest.

III. Statistical Support for Main Tables

A. Source-Level Significance

Table II gives the paired-bootstrap support for the first-stage source diagnostics summarized in the main paper. The confidence interval tests WHG-KV against the corresponding first-stage baseline. The final column reports the point-estimate margin between WHG-KV and the strongest non-WHG structured complement in the same dataset–first-stage block.

The source-level diagnostics use seeds 13, 29, and 43, 5,000 bootstrap samples per dataset–first-stage row, frozen train/dev prediction files, and no new LLM extraction calls. The artifact README records the ColBERTv2, BM25, Contriever, and HotpotQA source-level result roots, hashes, and redistributability notes.

B. Dense-Aware Controller Significance

The main paper summarizes the dense-aware controller bootstrap audit in prose. Table III reports the underlying paired-bootstrap F1 differences against ColBERTv2 and learned HKVM calibration, and Table IV reports the corresponding EM differences. The controller improves both F1 and EM over both comparators on all three benchmarks. The 2Wiki gain over learned HKVM is intentionally small but positive, because learned HKVM calibration and the HKVM-only controller are already strong on that dataset.

Controller bootstrap intervals use 5,000 samples over frozen prediction files with out-of-fold HKVM training signals. The source-level HotpotQA ColBERTv2 row in Table II comes from a separate source-replacement diagnostic, which explains its slightly different confidence interval from the main controller-vs-ColBERTv2 interval.

IV. Evidence-Selection and Standalone Boundaries

A. Deterministic Held-Out Slice Audit

To check whether the controller gains are concentrated in a small subset of the development predictions, we run an additional audit over frozen dense-aware controller prediction files.

TABLE I
SUPPLEMENTARY EVIDENCE MAP AND CLAIM BOUNDARY. THIS NAVIGATION TABLE MAPS EACH EVIDENCE BLOCK TO THE PAPER-FACING CLAIM IT SUPPORTS AND TO THE STRONGER INTERPRETATIONS IT DOES NOT SUPPORT; NUMERICAL EVIDENCE APPEARS IN THE FOLLOWING SECTIONS.

Evidence block	Supports	Excluded interpretation
Fixed-substrate protocol	Shared candidates, tuple caches, extraction scope, seeds, and manifests	Cross-extractor, cross-reader, or deployment generalization
Statistical support	Controller gains and WHG-specific source diagnostics	Official hidden-test significance or production traffic evidence
Evidence selection	Oracle headroom, learned recovery path, and HotpotQA standalone boundary	Deployable oracle systems or standalone WHG-KV superiority on HotpotQA
Feature/source mechanism	Dense–HKVM rank-score calibration; limits of naive fusion and interaction-only features	Arbitrary structured-source gains or handcrafted interactions as the core mechanism
Transfer and calibration	Leave-one-target transfer, feature ablation, target calibration, and subset stability	Target-data independence, monotonic sample efficiency, or hidden-test generalization
Parameters and baselines	Hyperparameter sensitivity, extraction boundary, and enhanced pairwise KG baselines	LLM triples dominating all OpenIE variants or tuning fixing all standalone failures
Cost and reproducibility	Fixed-output scoring cost, controller replay, artifact footprint, and provenance	Online serving throughput, fresh extraction latency, or unrestricted provider-artifact release

TABLE II
SOURCE-LEVEL BASELINE-VS-WHG PAIRED-BOOTSTRAP SUPPORT AND NON-WHG MARGINS. THE CONFIDENCE INTERVAL TESTS WHG-KV AGAINST THE CORRESPONDING FIRST-STAGE BASELINE; THE FINAL COLUMN IS A POINT-ESTIMATE MARGIN OVER THE STRONGEST MATCHED NON-WHG SOURCE. THE TABLE DOES NOT IMPLY THAT ALL STRUCTURED SOURCES HELP OR THAT THE DIAGNOSTICS COVER ALL RETRIEVER FAMILIES.

Dataset	First-stage	WHG-KV ΔF1 vs base	95% CI	Margin over best non-WHG
2WikiMultiHopQA	ColBERTv2	$+11.084$	$[+10.722, +11.442]$	$+8.077$
2WikiMultiHopQA	BM25	$+16.452$	$[+16.039, +16.865]$	$+9.786$
2WikiMultiHopQA	Contriever	$+23.119$	$[+22.668, +23.577]$	$+8.845$
MuSiQue	ColBERTv2	$+6.763$	$[+5.813, +7.753]$	$+8.219$
MuSiQue	BM25	$+12.038$	$[+10.935, +13.120]$	$+12.212$
MuSiQue	Contriever	$+7.813$	$[+6.810, +8.814]$	$+7.605$
HotpotQA	ColBERTv2	$+5.966$	$[+5.479, +6.456]$	$+5.397$
HotpotQA	BM25	$+4.971$	$[+4.554, +5.374]$	$+4.188$
HotpotQA	Contriever	$+5.166$	$[+4.683, +5.634]$	$+5.522$

TABLE III
DENSE-AWARE CONTROLLER F1 SIGNIFICANCE. DIFFERENCES ARE DENSE-AWARE HKVM CONTROLLER MINUS THE LISTED COMPARATOR. CONFIDENCE INTERVALS ARE 5,000-SAMPLE PAIRED-BOOTSTRAP INTERVALS OVER FROZEN PREDICTION FILES.

Dataset	Comparator	ΔF1	95% CI
2Wiki	ColBERTv2	$+11.084$	$[+10.740, +11.437]$
2Wiki	Learned HKVM	$+0.455$	$[+0.368, +0.542]$
MuSiQue	ColBERTv2	$+6.763$	$[+5.791, +7.726]$
MuSiQue	Learned HKVM	$+8.319$	$[+7.327, +9.293]$
HotpotQA	ColBERTv2	$+5.966$	$[+5.491, +6.452]$
HotpotQA	Learned HKVM	$+2.881$	$[+2.555, +3.201]$

TABLE IV
DENSE-AWARE CONTROLLER EM SIGNIFICANCE. DIFFERENCES ARE DENSE-AWARE HKVM CONTROLLER MINUS THE LISTED COMPARATOR.

Dataset	Comparator	ΔEM	95% CI
2Wiki	ColBERTv2	$+11.463$	$[+11.116, +11.820]$
2Wiki	Learned HKVM	$+0.477$	$[+0.388, +0.565]$
MuSiQue	ColBERTv2	$+6.964$	$[+5.984, +7.937]$
MuSiQue	Learned HKVM	$+8.509$	$[+7.509, +9.492]$
HotpotQA	ColBERTv2	$+5.991$	$[+5.510, +6.482]$
HotpotQA	Learned HKVM	$+2.921$	$[+2.579, +3.255]$

The audit applies a deterministic SHA-256 split over example ids and holds out 30% of the aligned frozen prediction rows. It then recomputes metrics and paired-bootstrap differences from saved per-example metrics only. No controller is trained, no retrieval or answer generation is rerun, and no LLM extraction calls are made. Table V reports the held-out F1 results.

The held-out deltas are consistent with the full-dev controller deltas: $+7.251$ vs $+6.763$ F1 on MuSiQue, $+11.196$ vs $+11.084$ on 2WikiMultiHopQA, and $+5.771$ vs $+5.966$ on HotpotQA when compared with ColBERTv2. All three seed-level held-out F1 deltas are positive for both ColBERTv2 and learned HKVM comparators. This audit reduces the concern that the dense-aware controller result is driven by a small unstable slice of the frozen dev predictions, but it does not change the evaluation boundary: the result remains a development-prediction robustness audit, not an official hidden-test result. A true train-internal held-out rerun was not attempted in this artifact pass because the old train-side dense-prediction inputs and merged train-subset files recorded in the historical manifests are no longer present in the working run directory.

B. Evidence-Selection Trajectory

Table VI records the trajectory from no-gold WHG retrieval to oracle answer-path selection, train-derived support scoring,

TABLE V

DETERMINISTIC HELD-OUT SLICE AUDIT OVER FROZEN DENSE-AWARE CONTROLLER DEV PREDICTIONS. THE SPLIT IS AN AUDIT SLICE OF SAVED PREDICTIONS, NOT A HIDDEN-TEST BENCHMARK. CONFIDENCE INTERVALS ARE 5,000-SAMPLE MATCHED-EXAMPLE BOOTSTRAP INTERVALS ON THE HELD-OUT SLICE.

Dataset	Held-out ex./seed	Controller F1	ΔF1 vs ColBERTv2	95% CI	ΔF1 vs learned HKVM	95% CI
MuSiQue	746	64.214	$+7.251$	$[+5.423, +9.109]$	$+8.373$	$[+6.689, +10.138]$
2WikiMultiHopQA	3,863	88.959	$+11.196$	$[+10.558, +11.837]$	$+0.454$	$[+0.302, +0.613]$
HotpotQA	2,227	84.377	$+5.771$	$[+4.909, +6.639]$	$+3.351$	$[+2.741, +3.958]$

learned HKVM calibration, and the dense-aware controller. This table is a diagnostic trajectory rather than a single deployed pipeline. The oracle rows use support information and serve only to measure headroom.

The trajectory combines fixed-substrate dev outputs, oracle diagnostics, train-to-dev support-scoring outputs, learned HKVM calibration, and dense-aware controller predictions. It should be read as a mechanism-oriented evidence path rather than as a single chronological production pipeline.

C. Standalone Boundary on HotpotQA

HotpotQA is included to prevent a standalone-retrieval overclaim. In the fixed-substrate standalone matrix, WHG-KV improves support coverage over KG-PPR but does not improve answer F1. Table VII separates that standalone boundary from the controller-mediated recovery reported in the main paper.

This boundary is compatible with the dense-aware controller result: raw structured retrieval can underperform lexical and dense baselines on HotpotQA, while WHG-KV rank/score signals can still help a learned controller reorder answer-bearing passages.

V. SOURCE SPECIFICITY AND FEATURE MECHANISM

A. Naive Fusion

MuSiQue contains genuine ColBERTv2–WHG complementarity, but simple fixed-budget fusion and shallow single-feature gates did not reliably exceed ColBERTv2. RRF fusion of ColBERTv2 and HKVM predictions improved F1 over ColBERTv2 by only +0.110 on MuSiQue, far below the learned controller's +6.763. On 2WikiMultiHopQA and HotpotQA, RRF improved by +8.800 and +3.379 F1 respectively, but remained below the learned controller (+11.084 and +5.966). Three additional fixed-budget quota mixtures (dense3+hkvm2, dense4+hkvm1) were evaluated on MuSiQue and produced F1 values of 58.237, 58.309, and 57.923, none reliably exceeding ColBERTv2 (58.309). This motivates the learned passage-level controller used in the main paper, and blocks the simpler claim that generic RRF or quota fusion is enough.

B. Controller Feature Ablation

The dense-aware controller uses per-passage feature vectors containing ColBERTv2 and HKVM rank/score signals. Table VIII decomposes the full feature set to identify which signal groups carry the controller gain. The ablation keeps the controller protocol, train-to-dev split, and hyperparameters fixed while varying the feature subset supplied to the logistic ranker.

The main finding is that explicit dense-HKVM interaction features are not necessary for the MuSiQue gain. The no-interaction variant reaches 65.738 F1 on MuSiQue, marginally exceeding the full controller (65.073). Score-only (65.254) also approaches full performance. Interaction-only (60.295) is far below full, confirming that interaction features alone are insufficient. On 2Wiki, HKVM calibration is already strong (88.391), so all dense+HKVM variants cluster within 0.7 F1 of each other. The safe mechanism claim is that dense and WHG-KV rank/score signals jointly carry useful passage-reordering information; explicit interaction features help on 2Wiki but are not the primary driver of the MuSiQue gain. This ablation uses out-of-fold HKVM training signals, seeds 13/29/43, 5,000-sample paired bootstrap, and no new LLM extraction calls.

VI. GENERALIZATION AND CALIBRATION BOUNDARY

The main paper visualizes the transfer/calibration audit as a compact boundary figure. This section gives the underlying tables and provenance. The audit first tests whether a controller trained without the target dataset transfers, then decomposes which feature sources carry that transfer, adds increasing amounts of target-domain training data, and finally repeats MuSiQue target subsets to check whether the deterministic-prefix anomaly is mainly a sampling artifact. All four analyses reuse saved ColBERTv2, learned-HKVM, RRF, and controller prediction files; they do not rerun extraction, retrieval, answer generation, or reader models.

A. Leave-One-Target Transfer

Table IX reports the leave-one-target pooled controller audit. The controller is trained on the two non-target datasets and evaluated on the held-out target benchmark. This is the cleanest transfer diagnostic because it avoids selecting a favorable source dataset after the fact.

Bootstrap diagnostics support the same bounded interpretation. Pooled transfer improves over ColBERTv2 with positive confidence intervals in all three seeds for all three targets. It also improves over learned HKVM in all seeds for MuSiQue and HotpotQA, but the 2Wiki learned-HKVM comparison is mixed because learned HKVM is already strong. Pooled transfer does not stably exceed target-trained controllers: 2Wiki is below target-trained in most seeds, MuSiQue is not stable, and HotpotQA is a near-tie with confidence intervals overlapping zero.

TABLE VI
EVIDENCE-SELECTION TRAJECTORY FROM NO-GOLD WHG RETRIEVAL TO DENSE-AWARE CONTROL. THE TABLE SUPPORTS THE INTERPRETATION THAT SUPPORT SELECTION IS A REPAIRABLE BOTTLENECK; ORACLE ROWS ARE DIAGNOSTIC UPPER BOUNDS AND NOT DEPLOYABLE MODEL RESULTS.

	2WikiMultiHopQA			MuSiQue			HotpotQA		
Variant	F1	EM	AR@10	F1	EM	AR@10	F1	EM	AR@10
No-gold WHG retrieval	79.299	78.650	99.650	39.925	39.222	38.643	72.957	72.586	98.190
Oracle answer-path selection	90.884	90.832	99.634	74.330	74.059	73.066	91.111	91.047	98.190
Train-derived support scorer	87.212	86.967	98.378	54.252	53.662	50.228	81.525	81.269	96.016
Learned HKVM calibration	88.391	88.181	99.030	56.754	56.213	53.303	82.929	82.705	96.439
Dense-aware HKVM controller	88.846	88.658	100.000	65.073	64.722	64.115	85.810	85.627	100.000

TABLE VII
HOTPOTQA STANDALONE BOUNDARY UNDER MATCHED CANDIDATE-PASSAGE BUDGETS. THE TABLE BLOCKS A STANDALONE WHG-KV SUPERIORITY CLAIM WHILE REMAINING COMPATIBLE WITH CONTROLLER-MEDIATED RECOVERY.

Method	AR@10	F1	EM
BM25	99.851	81.835	81.648
Contriever	100.000	79.930	79.730
ColBERTv2	100.000	79.844	79.635
KG-PPR	95.381	73.645	73.315
Static-HG	98.190	73.340	73.018
Weighted-HG-KV	98.190	72.957	72.586

TABLE VIII
DENSE-AWARE CONTROLLER FEATURE ABLATION. FULL USES BOTH SOURCE SIGNALS PLUS EXPLICIT DENSE-HKVM INTERACTION FEATURES; NO-INTERACTION DROPS THE EXPLICIT INTERACTION TERMS; SCORE-ONLY AND RANK-ONLY KEEP ONLY THOSE FEATURE GROUPS; INTERACTION-ONLY USES ONLY THE EXPLICIT CROSS-SOURCE FEATURES. COLBERTV2 AND LEARNED HKVM CALIBRATION ROWS ARE REFERENCE PREDICTORS INCLUDED IN THE MAIN PAPER.

Variant	MuSiQue F1	2Wiki F1
ColBERTv2 (reference)	58.309	77.763
Learned HKVM calibration (reference)	56.754	88.391
Controller dense-only	58.247	77.679
Controller HKVM-only	56.760	88.415
Controller interaction-only	60.295	88.769
Controller rank-only	64.474	88.637
Controller score-only	65.254	88.696
Controller no-interaction	65.738	88.581
Controller full	65.073	88.846

B. *Transfer Feature Source*

Table X decomposes the leave-one-target transfer controller. Dense-only transfer is insufficient on all targets; HKVM-only nearly matches full on 2Wiki but is insufficient on MuSiQue and HotpotQA. Explicit interaction features are not necessary: no-interaction, rank-only, score-only, or interaction-only variants can match the full controller by point estimate on some datasets. The transferable mechanism should therefore be described as learned calibration over dense and WHG-KV rank/score channels, not as a gain from explicit interaction terms.

Full-vs-ablation bootstrap gives three stable conclusions. First, full transfer is strictly above dense-only on all targets. Second, full transfer is strictly above HKVM-only on MuSiQue and HotpotQA, while 2Wiki is an HKVM-favorable boundary where HKVM-only is close to full. Third, interaction-only transfer is insufficient on MuSiQue and does not produce a stable improvement over full elsewhere. These signs are the basis for the main paper's narrowed mechanism wording.

C. *Target-Domain Calibration Curve*

The target-domain calibration audit starts from the pooled transfer controller and adds deterministic prefixes of target-dataset training examples. Table XI summarizes the best observed prefix point for each target. 2Wiki is saturated: pooled transfer is already within 0.186 F1 of target-trained, so the closure ratio is numerically unstable. HotpotQA shows the cleanest target-calibration benefit, reaching 53.7% gap closure at 250 examples and 79.7% at 10,000 examples. MuSiQue has the largest possible gain but a nonmonotonic prefix curve, motivating the repeated-subset stability audit.

The full prefix curve is reported in Table XII. The nonmonotonic rows are not a nuisance to hide: they block a sample-complexity claim and show that target calibration depends on dataset and sample composition.

D. *MuSiQue Target-Subset Stability*

The repeated-subset audit focuses on MuSiQue, the dataset with the strongest deterministic-prefix anomaly. Each target size has 30 runs, corresponding to 3 seeds and 10 subset repeats. Table XIII shows that larger random subsets become more reliable: mean F1 rises from 60.060 at 50 examples to 64.645 at 2,500 examples, while standard deviation drops from 6.465 to 1.090. At 2,500 examples, 27/30 runs beat their seed-specific pooled transfer baseline.

Table XIV compares deterministic prefixes against random-subset means. The largest correction is at 1,000 examples: the prefix result is 60.752 F1, but the random-subset mean is 63.964 F1. Thus the sharp deterministic-prefix nonmonotonicity is partly a prefix-sampling artifact. The audit still does not justify a monotonic sample-complexity claim, because small subsets remain unstable and the repeated-subset audit was run only for MuSiQue.

The resulting paper-safe synthesis is narrow but useful: dense/HKVM rank-score calibration transfers across the tested benchmarks under frozen prediction artifacts; the transferable signal benefits from target-domain calibration when enough target examples are available; and the benefit is dataset- and sample-composition-sensitive. The experiments do not

TABLE IX
LEAVE-ONE-TARGET POOLED TRANSFER OVER FROZEN PREDICTION ARTIFACTS. THE POOLED CONTROLLER IS TRAINED WITHOUT THE TARGET DATASET. IT IMPROVES OVER COLBERTV2 AND RRF ON ALL THREE TARGETS, WHILE TARGET-TRAINED CONTROLLERS REMAIN STRONGEST OR NEAR-STRONGEST.

Target	ColBERTv2	Learned HKVM	RRF	Pooled transfer	Target-trained	Δ vs ColBERTv2	Δ vs HKVM	Δ vs target
2WikiMultiHopQA	77.763	88.391	86.563	88.555	88.846	+10.793	+0.164	-0.291
MuSiQue	58.309	56.754	58.420	63.645	65.073	+5.335	+6.891	-1.428
HotpotQA	79.844	82.929	83.223	85.706	85.810	+5.862	+2.777	-0.104

TABLE X
TRANSFER FEATURE-SOURCE ABLATION. VALUES ARE MEAN F1 UNDER THE LEAVE-ONE-TARGET POOLED-TRANSFER PROTOCOL. POSITIVE "FULL–VARIANT" VALUES MEAN THE FULL DENSE+HKVM CONTROLLER OUTPERFORMS THE ABLATED FEATURE SET.

Variant	2Wiki F1	Full–variant	MuSiQue F1	Full–variant	HotpotQA F1	Full–variant
Full dense+HKVM	88.555	0.000	63.645	0.000	85.706	0.000
Dense only	77.270	+11.285	57.657	+5.988	79.844	+5.862
HKVM only	88.295	+0.260	56.754	+6.891	82.929	+2.777
No interaction	88.777	-0.222	61.730	+1.914	85.341	+0.365
Rank only	88.539	+0.016	61.904	+1.741	85.626	+0.080
Score only	86.567	+1.988	64.235	-0.591	84.787	+0.919
Interaction only	87.967	+0.588	59.247	+4.398	85.652	+0.054

TABLE XI
BEST OBSERVED TARGET-DOMAIN CALIBRATION UNDER DETERMINISTIC-PREFIX SAMPLING. GAP CLOSURE IS MEASURED FROM POOLED TRANSFER TOWARD THE TARGET-TRAINED CONTROLLER.

Target	Best n	Best F1	Pooled F1	Target F1	Closure
2Wiki	10000	88.753	88.660	88.846	0.498
HotpotQA	10000	85.689	85.216	85.810	0.797
MuSiQue	500	64.914	63.838	65.073	0.872

support hidden-test generalization, target-data independence, or monotonic sample efficiency.

VII. PARAMETER AND BASELINE ROBUSTNESS

A. Extraction Substrate

The shared LLM-triple cache is a controlled extraction substrate, not a claim that LLM triples dominate all OpenIE alternatives. In extraction-quality diagnostics, LLM triples were much sparser than heuristic OpenIE: objects per passage changed from 34.814 to 4.249 on 2Wiki and from 47.056 to 4.878 on MuSiQue. Pairwise KG-PPR with the sparse LLM tuple cache did not dominate heuristic OpenIE KG-PPR in that diagnostic. The paper therefore treats cache changes as new manifests rather than mixing them into the frozen result matrix.

B. Hyperparameter Sensitivity

Table XV reports WHG-KV sensitivity to the three main hyperparameter groups: the confidence-weight coefficients $\lambda = (\lambda_f, \lambda_s, \lambda_b)$, the weight-scale parameter β, and the diffusion parameter ρ. Each variant varies one parameter while holding the others at their default values (λ_f=0.20, λ_s=0.40, λ_b=0.40, β=2.0, ρ=0.35). All runs reuse the existing DeepSeek evidence tuple cache and evaluate on the MuSiQue dev answerable subset with seed 13.

Across all 14 variants, WHG-KV F1 spans 38.471–39.918 (Δ=1.447), and all variants exceed the pairwise KG-PPR baseline (36.332 F1, Table 1 in the main paper). The default configuration is conservative: higher β and lower ρ modestly improve F1, but the paper's qualitative conclusions are robust to all tested hyperparameter perturbations. The default was chosen to avoid dev-set over-tuning rather than to maximize single-seed MuSiQue F1.

C. Enhanced Pairwise KG Baselines

To determine whether the WHG-KV advantage is an artifact of a weak pairwise reference, we implement two topological enhancements from HippoRAG 2 [1] within the same fixed substrate. The first adds passage nodes: each passage becomes a graph vertex connected to the entity nodes it contains, allowing PPR to flow through passage-level context. The second adds query-to-triple linking: PPR seeds are drawn from query–triple token overlap in addition to entity matching. Both enhancements change only the graph topology; the evidence tuple cache, candidate passages, and retrieval algorithm (PPR) remain identical to the original KG-PPR baseline. Table XVI reports four variant combinations across all three benchmarks.

Passage nodes alone shift F1 by less than 0.7 points relative to the original pairwise baselines across all three datasets. Query-to-triple linking improves 2WikiMultiHopQA (+2.1 F1) but degrades MuSiQue (-1.3 to -1.5 F1), where token-overlap matching against compositional questions introduces noise. The strongest enhanced pairwise variant reaches 78.041 F1 on 2WikiMultiHopQA, which remains below WHG-KV (79.299). On MuSiQue the smallest gap is $+3.2$ F1. These results indicate that the hypergraph advantage is not an artifact of a weak pairwise reference: even the best pairwise topology improvements available do not reproduce the evidence-organization gain of answer-path hyperedges under the shared extraction substrate.

TABLE XII

TARGET-DOMAIN CALIBRATION CURVE. $n = 0$ IS THE LEAVE-ONE-TARGET POOLED-TRANSFER REFERENCE. THE CURVE IS AN AUDIT OVER DETERMINISTIC TARGET-PREFIX SAMPLES, NOT A REPEATED-RANDOM SAMPLE-COMPLEXITY ESTIMATE.

Target	Target examples	F1	Δ vs pooled	Δ vs target-trained	Gap closure
2Wiki	0	88.660	+0.000	-0.186	0.000
2Wiki	50	88.606	-0.053	-0.240	-0.286
2Wiki	100	88.531	-0.129	-0.316	-0.693
2Wiki	250	88.543	-0.117	-0.303	-0.626
2Wiki	500	88.752	+0.093	-0.094	0.496
2Wiki	1000	88.339	-0.320	-0.507	-1.719
2Wiki	2500	88.657	-0.003	-0.189	-0.016
2Wiki	5000	88.629	-0.031	-0.217	-0.166
2Wiki	10000	88.753	+0.093	-0.094	0.498
HotpotQA	0	85.216	+0.000	-0.594	0.000
HotpotQA	50	85.488	+0.272	-0.322	0.458
HotpotQA	100	85.481	+0.264	-0.329	0.445
HotpotQA	250	85.535	+0.319	-0.275	0.537
HotpotQA	500	85.295	+0.079	-0.515	0.133
HotpotQA	1000	85.455	+0.238	-0.355	0.401
HotpotQA	2500	85.419	+0.202	-0.391	0.341
HotpotQA	5000	85.378	+0.161	-0.432	0.272
HotpotQA	10000	85.689	+0.473	-0.121	0.797
MuSiQue	0	63.838	+0.000	-1.235	0.000
MuSiQue	50	62.491	-1.347	-2.582	-1.091
MuSiQue	100	60.862	-2.976	-4.211	-2.409
MuSiQue	250	60.493	-3.344	-4.580	-2.708
MuSiQue	500	64.914	+1.076	-0.159	0.872
MuSiQue	1000	60.752	-3.086	-4.321	-2.498
MuSiQue	2500	64.782	+0.944	-0.291	0.764
MuSiQue	5000	63.488	-0.350	-1.585	-0.283
MuSiQue	10000	63.905	+0.067	-1.168	0.054

TABLE XIII

MUSIQUE REPEATED RANDOM TARGET-SUBSET AUDIT. "POSITIVE/NEGATIVE" COUNTS COMPARE EACH RUN AGAINST ITS SEED-SPECIFIC POOLED-TRANSFER BASELINE.

Target examples	Runs	F1 mean	F1 std	F1 min	F1 median	F1 max	Mean Δ vs pooled	Pos./neg.
50	30	60.060	6.465	37.396	62.411	62.919	-3.778	0/30
100	30	61.548	2.850	47.158	62.283	63.282	-2.289	0/30
250	30	61.963	2.549	54.330	62.560	65.260	-1.875	5/25
500	30	63.258	2.309	53.971	64.315	65.653	-0.579	16/14
1000	30	63.964	1.499	58.878	64.385	65.578	+0.126	21/9
2500	30	64.645	1.090	60.209	64.956	65.834	+0.807	27/3

TABLE XIV

DETERMINISTIC PREFIXES VERSUS RANDOM-SUBSET MEANS ON MUSIQUE. POSITIVE DIFFERENCES MEAN THE RANDOM-SUBSET MEAN IS ABOVE THE PREFIX RESULT.

Target examples	Prefix F1	Random mean	Difference
50	62.491	60.060	-2.431
100	60.862	61.548	+0.686
250	60.493	61.963	+1.470
500	64.914	63.258	-1.656
1000	60.752	63.964	+3.212
2500	64.782	64.645	-0.137

VIII. COST AND RESOURCE DISCLOSURE

The paper-facing artifact separates one-time extraction cost from frozen-output verification. The main structured matrix uses 202,176 cached dev passage-level extractor inputs: 125,760 for 2WikiMultiHopQA, 48,325 for MuSiQue, and 28,091 for HotpotQA. The train-side controller inputs use 399,491 additional cached passage-level extractor inputs. These counts are extraction-scope counts, not retrieval-budget counts; retrieval and answer assembly still use the shared top-10 retrieval and top-5 answer-context protocol described in the main paper. All supplementary tables are computed from frozen prediction files and therefore introduce no new LLM extraction calls.

The released artifact records the resource footprint needed to verify the submitted evidence. In the staged release manifest, benchmark data occupy 355,293,085 bytes, LLM extraction caches occupy 734,269,603 bytes, frozen prediction outputs occupy 1,200,203,334 bytes, and compact result summaries occupy 1,538,964 bytes. The package also includes a SHA-

TABLE XV
WHG-KV HYPERPARAMETER SENSITIVITY ON MUSIQUE (SEED 13). DEFAULT ROW USES THE PAPER-FACING CONFIGURATION. ALL VARIANTS REUSE THE FROZEN DEEPSEEK EVIDENCE TUPLE CACHE; F1 VALUES ARE SINGLE-SEED POINT ESTIMATES.

Group	Variant	AR@10	F1	EM
Default				
	Paper default	37.815	38.946	38.229
β *(weight scale)*				
	$\beta = 1.5$	37.360	38.693	37.940
	$\beta = 2.5$	38.395	39.482	38.767
	$\beta = 3.0$	38.395	39.834	39.139
	$\beta = 4.0$	39.015	39.918	39.222
ρ *(diffusion)*				
	$\rho = 0.15$	38.477	39.383	38.726
	$\rho = 0.25$	37.857	39.292	38.602
	$\rho = 0.45$	37.650	38.937	38.146
	$\rho = 0.55$	36.947	38.471	37.650
λ *(confidence weights)*				
	Uniform $(0.33, 0.33, 0.33)$	37.940	38.988	38.271
	Factual-only $(1, 0, 0)$	38.188	39.200	38.477
	Salience-only $(0, 1, 0)$	37.898	38.936	38.229
	Bridge-only $(0, 0, 1)$	37.940	39.022	38.271
	Factual+Bridge $(0.5, 0, 0.5)$	38.022	38.943	38.229

TABLE XVI
ENHANCED PAIRWISE KG-PPR BASELINES WITH PASSAGE NODES (PN) AND QUERY-TO-TRIPLE LINKING (QT). ALL VARIANTS REUSE THE SHARED DEEPSEEK EVIDENCE TUPLE CACHE; NO NEW LLM EXTRACTION CALLS. WHG-KV FROM THE MAIN PAPER IS INCLUDED FOR REFERENCE.

Method	2Wiki F1	MuSiQue F1	HotpotQA F1
KG-PPR (original)	75.873	36.332	73.645
Weighted-KG-PPR (original)	75.894	36.378	73.695
KG-PPR + PN	75.414	36.710	72.988
Weighted-KG-PPR + PN	75.746	36.460	73.360
KG-PPR + PN + QT	78.015	35.052	71.972
Weighted-KG-PPR + PN + QT	78.041	34.803	71.997
WHG-KV (reference)	79.299	39.925	72.957

256 file manifest, reproduction scripts, run summaries, and per-family provenance records. Dense model weights and API credentials are not redistributed; rerunning dense retrieval requires downloading the upstream model weights, and changing the LLM endpoint, prompt, schema, or extraction budget requires regenerating the extraction caches and recording a new cache manifest.

Table XVII reports a read-only cost/efficiency audit over the submitted evidence artifacts. The Weighted-HG-KV and ColBERTv2 columns are per-example scoring latencies recorded in the fixed-substrate run summaries. The controller-replay column measures a deterministic CPU-side replay of saved dense-aware controller models over frozen ColBERTv2 and learned-HKVM prediction files, using seven timing repeats after one warm-up repeat. The audit completed with zero LLM API calls, zero training runs, zero retrieval reruns, and zero answer-generation reruns.

These diagnostics are bounded cost measurements for the fixed-substrate artifact, not deployment-throughput claims. They do not include full-corpus indexing, dense model loading, online serving queues, reader-model latency, or fresh LLM extraction latency. The experiments evaluate evidence organization over fixed candidate passages rather than a production RAG service, so the reported timings, artifact footprint, and cache manifests should be read as reproducibility and audit costs for the submitted evidence matrix.

IX. RUN AND ARTIFACT PROVENANCE

The public artifact is split across two hosting platforms:

- **GitHub** (code, configs, scripts, results, manifests): https://github.com/Mingyu-Zh/HKVM-RAG
- **Hugging Face Dataset** (data, frozen outputs): https://huggingface.co/datasets/MingY-Zh/HKVM-RAG

This supplement itself is hosted in the GitHub repository at `supplemental_material.pdf`. Reviewers should download `data/` and `frozen_outputs/` from Hugging Face and place them at the repository root before running the paper-evidence verification script.

Table XVIII lists the paper-facing result families and the artifact records required to verify them. The artifact includes scripts (`scripts/verify_paper_evidence.py`, `scripts/reproduce_fixed_substrate.py`), configs (`configs/`), and a SHA-256 file manifest (`manifests/FILES.sha256`) for integrity verification.

REFERENCES

[1] B. J. Gutierrez, Y. Shu, W. Qi, S. Zhou, and Y. Su, "From RAG to memory: Non-parametric continual learning for large language models," in *Proceedings of the 42nd International Conference on Machine Learning*, PMLR, vol. 267, pp. 21497–21515, 2025.

TABLE XVII

FIXED-SUBSTRATE COST/EFFICIENCY AUDIT. LATENCIES ARE MILLISECONDS PER EXAMPLE OVER FROZEN CANDIDATE-PASSAGE ARTIFACTS. CONTROLLER REPLAY MEASURES FEATURE ALIGNMENT, SCORING, AND RERANKING OVER SAVED PREDICTION FILES; IT IS NOT A DEPLOYMENT-THROUGHPUT BENCHMARK.

Dataset	Ex./seed	Cand./seed	Weighted-HG-KV ms/ex.	ColBERTv2 ms/ex.	Replay ms/ex.
MuSiQue	2,417	33,940	0.330	53.190	0.288
2WikiMultiHopQA	12,576	125,237	0.157	29.771	0.207
HotpotQA	7,405	73,700	0.264	61.689	0.200

TABLE XVIII

PAPER-FACING RESULT PROVENANCE. THE TABLE IDENTIFIES THE PROTOCOL AND ARTIFACT RECORD FOR EACH RESULT BLOCK; EXACT RESULT-FILE ROOTS, HASHES, AND REDISTRIBUTABILITY CONSTRAINTS BELONG IN THE SUBMITTED ARTIFACT README.

Result block	Verifies	Prediction protocol	Statistical support	Artifact record
Fixed-substrate structured retrieval matrix	Standalone WHG-vs-KG comparison and HotpotQA boundary	Frozen dev predictions; shared tuple cache; seeds 13/29/43	Seed-13 1,000-sample WHG-minus-KG bootstrap; claim-specific table files	Fixed-substrate result files and frozen-output manifest
Evidence-selection trajectory	Oracle headroom, support scorer, calibration, and controller sequence	Frozen dev predictions plus train-to-dev models; no dev labels at inference	Claim-specific result files; controller rows use 5,000-sample bootstrap	Calibration-analysis result files and run summaries
Dense-aware controller	Dense-aware controller vs ColBERTv2 and learned HKVM	OOF train-side HKVM predictions; frozen dev predictions; seeds 13/29/43	5,000-sample paired bootstrap over frozen prediction files	Dense-aware controller result files and frozen-output manifest
Transfer and calibration boundary	Cross-dataset transfer, feature-source narrowing, target calibration, and MuSiQue subset stability	Frozen train/dev prediction artifacts; no new extraction, retrieval, or answer generation; seeds 13/29/43	1,000-sample bootstrap for transfer/calibration rows; 30 MuSiQue random-subset runs per reported target size	Transfer/calibration provenance records
Held-out slice audit	Robustness of controller gains on a deterministic 30% audit slice	Frozen dense-aware controller dev predictions; no retraining or reranking	5,000-sample matched-example bootstrap on held-out rows	Held-out audit result files and manifest
Cost/efficiency audit	Runtime/resource boundary for the fixed-substrate artifact	Frozen controller predictions, fixed-substrate summaries, and release manifest; no LLM calls	Seven-repeat controller replay plus run-summary latency records	Cost/replay audit result files and manifest
Source-level diagnostics	WHG-KV source specificity under ColBERTv2/BM25/Contriever	First-stage source fixed per block; structured source swapped; no new LLM calls	5,000-sample baseline-vs-WHG bootstrap; non-WHG margins are point estimates	Source-level ablation result files and bootstrap CSVs
Negative diagnostics	Naive-fusion and extraction-substrate boundaries	Diagnostic result files outside the frozen main matrix	Descriptive unless explicitly reported as paired bootstrap	Extraction-quality summaries and prompt/schema records